Geometrical representation of flexibility

Polytope visualizations

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Simple flexibility example

- Battery system, 10KWh, max charge/discharge 5 KW

$$E_0 = 5.0$$
$$0 \leq E_1 \leq 10$$
$$0 \leq E_2 \leq 10$$
$$-5 \leq P_1 \leq 5$$
$$-5 \leq P_2 \leq 5$$

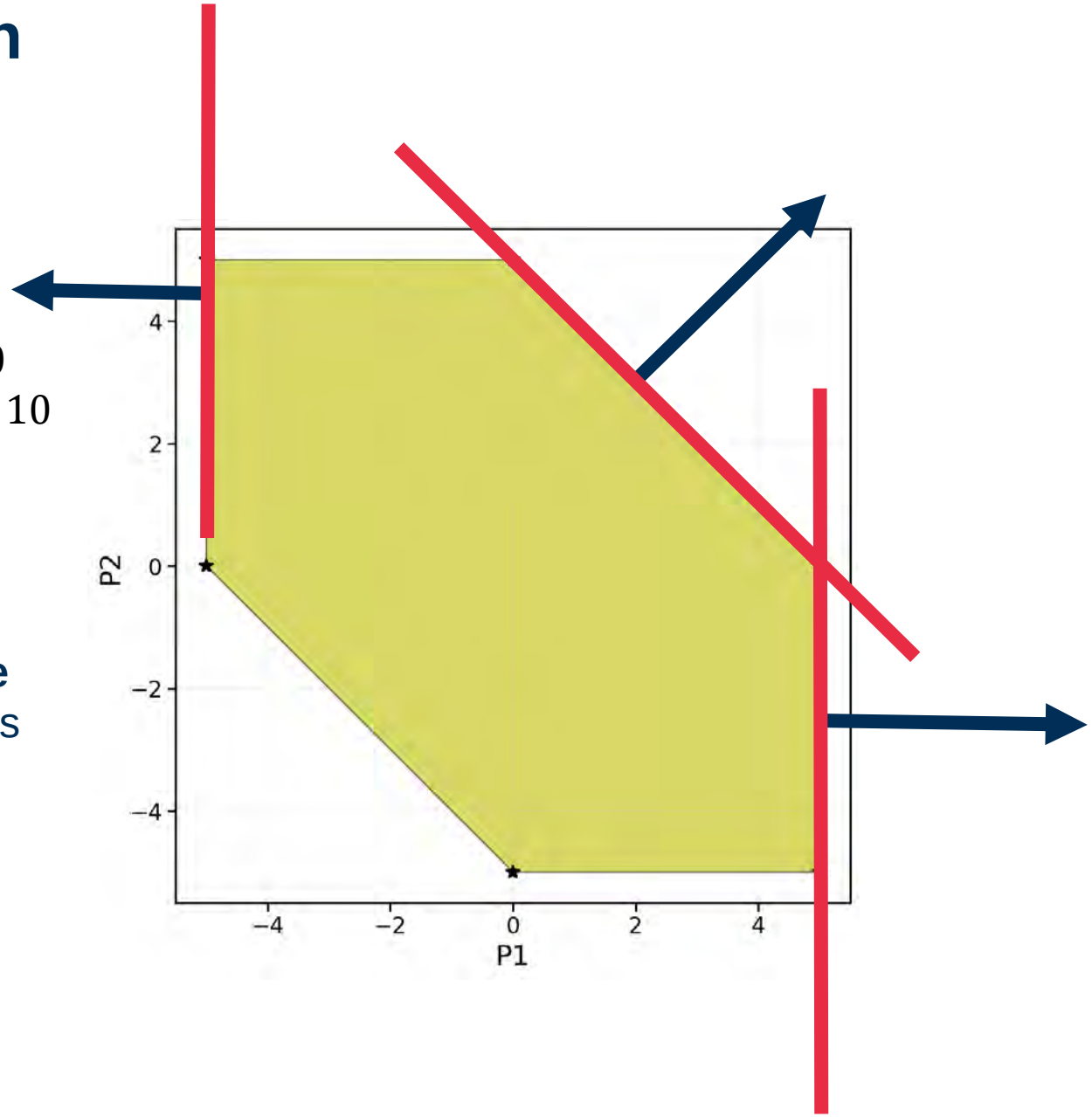
- Eliminate variables E

$$E_1 = E_0 + P_1$$
$$E_2 = E_1 + P_2 = E_0 + P_1 + P_2$$

2D visualization

On P space

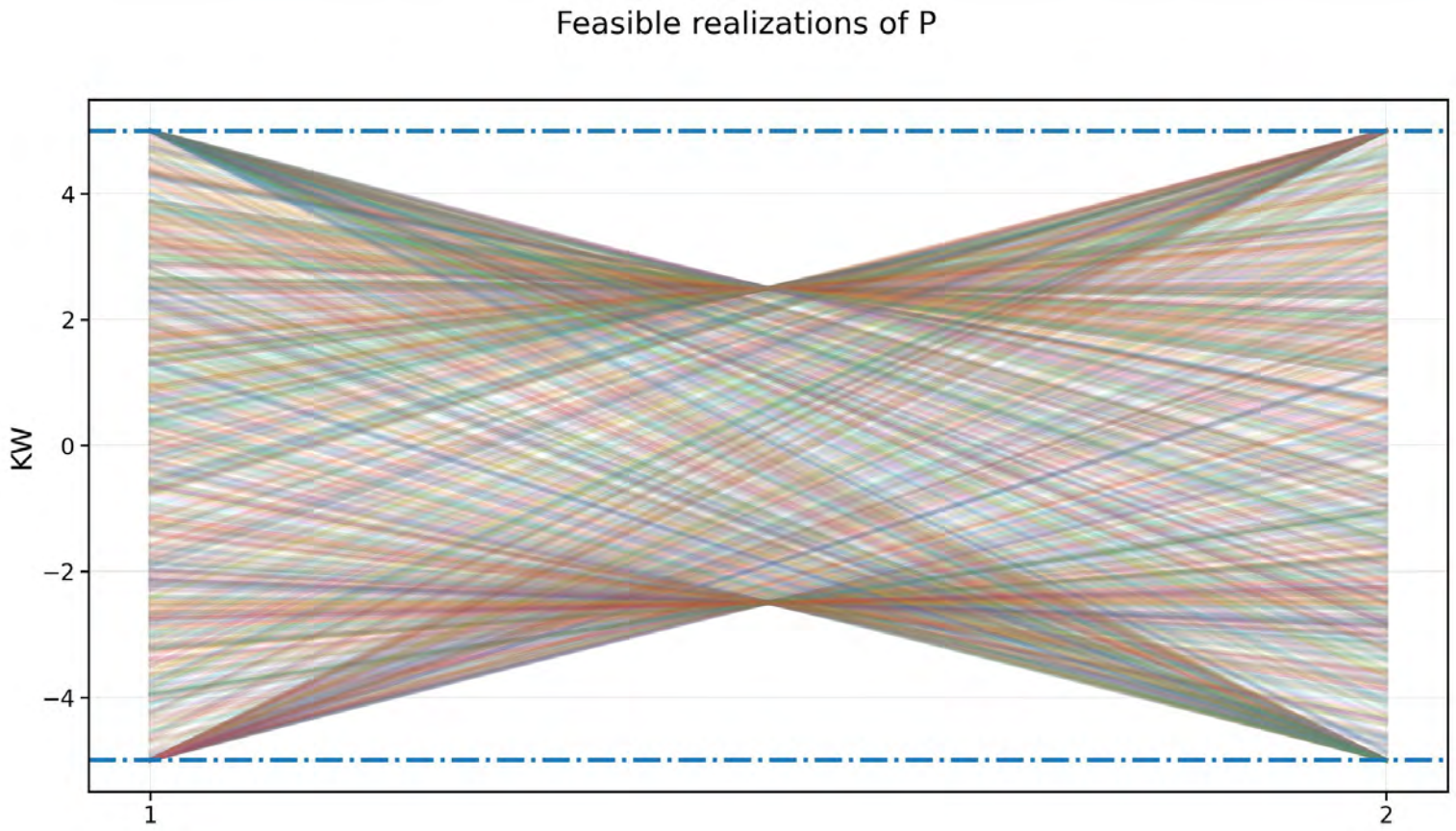
$$E_0 = 5.0$$
$$0 \leq E_0 + P_1 \leq 10$$
$$0 \leq E_0 + P_1 + P_2 \leq 10$$
$$-5 \leq P_1 \leq 5$$
$$-5 \leq P_2 \leq 5$$



We are visualizing the **feasible set** of the system of inequalities

Time slot plot

On P space



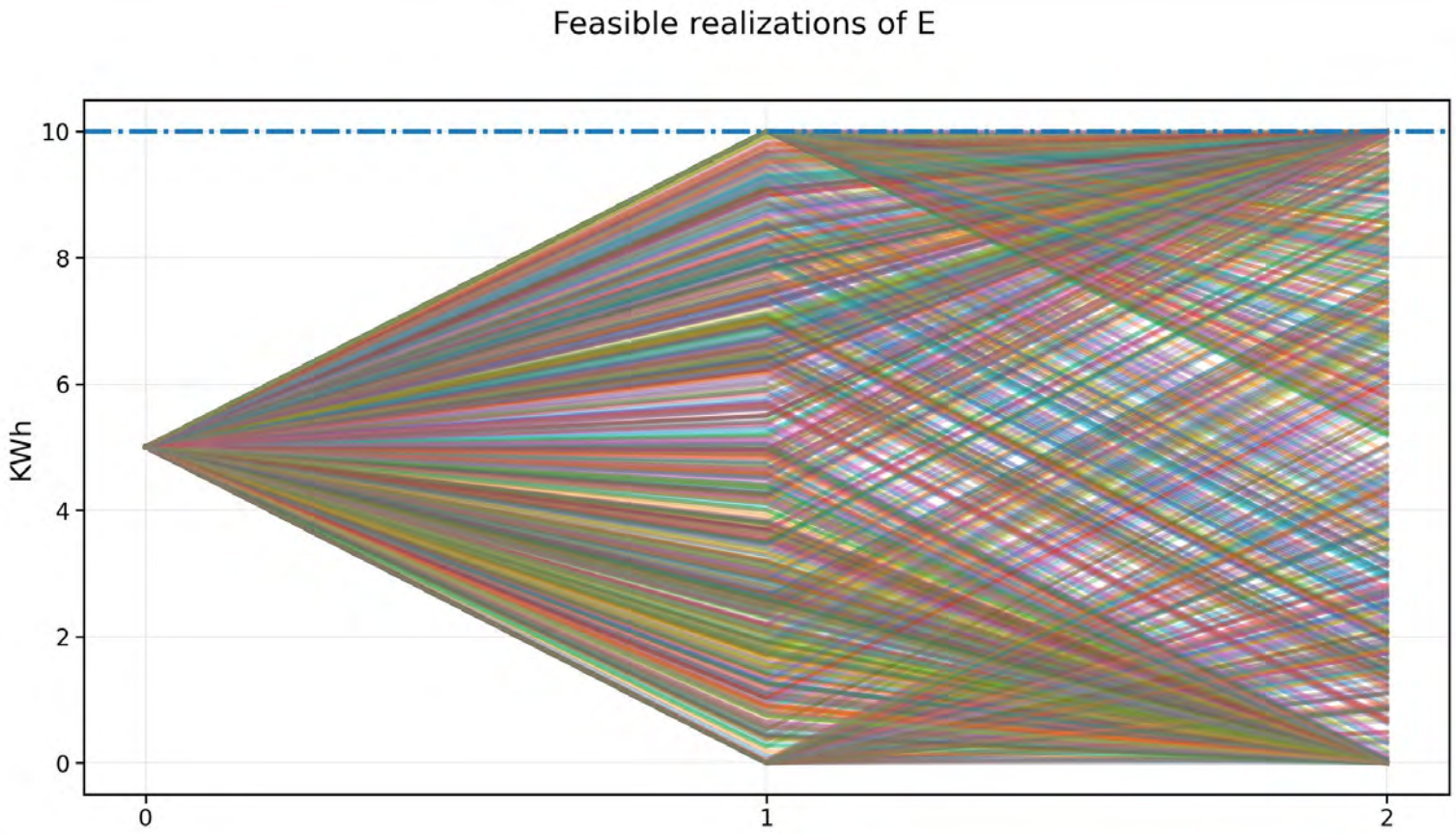
- You can always find back E (charge state)

$$\boldsymbol{E} = E_0 + \begin{bmatrix} 1 & 0 & 0 \\ \vdots & \ddots & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} P_1 \\ \vdots \\ P_n \end{bmatrix}$$

$$\boldsymbol{E} = E_0 + \boldsymbol{LP}$$

Time slot plot

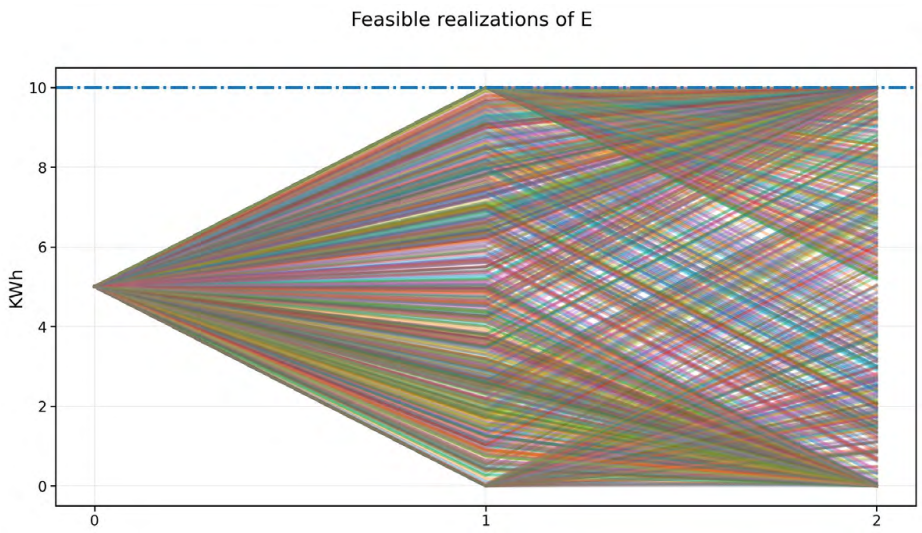
On E space

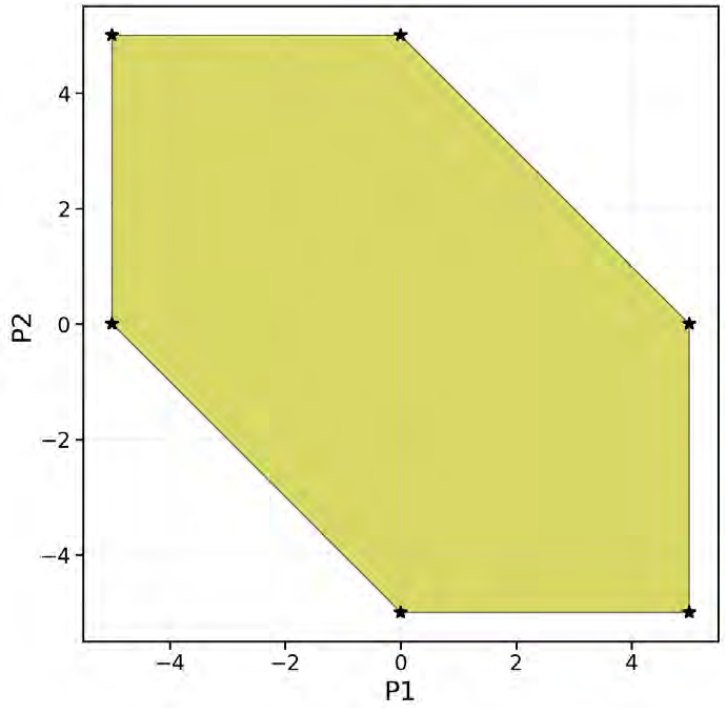


Time slot plot

On E space

- Flexibility is about having options
- We can operate this battery in many possible ways
- Feasible set representation



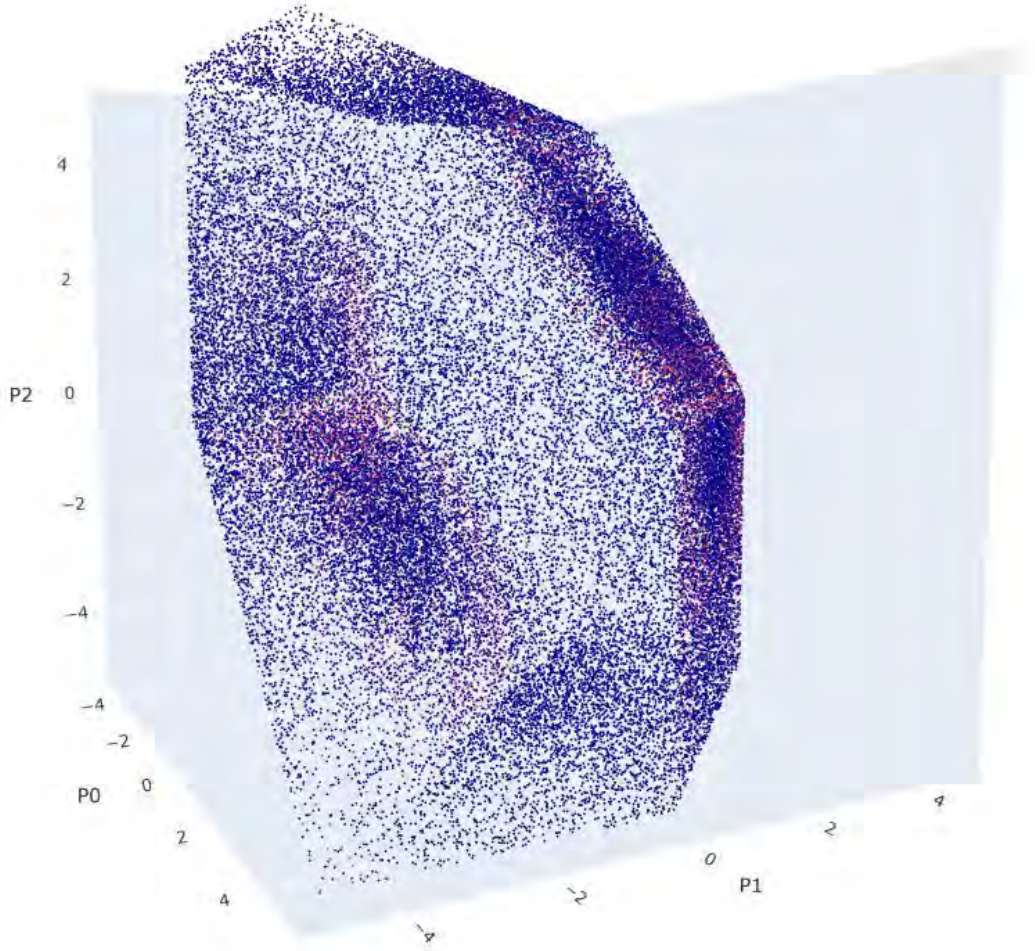


Simple flexibility example

- Let’s add more time slots

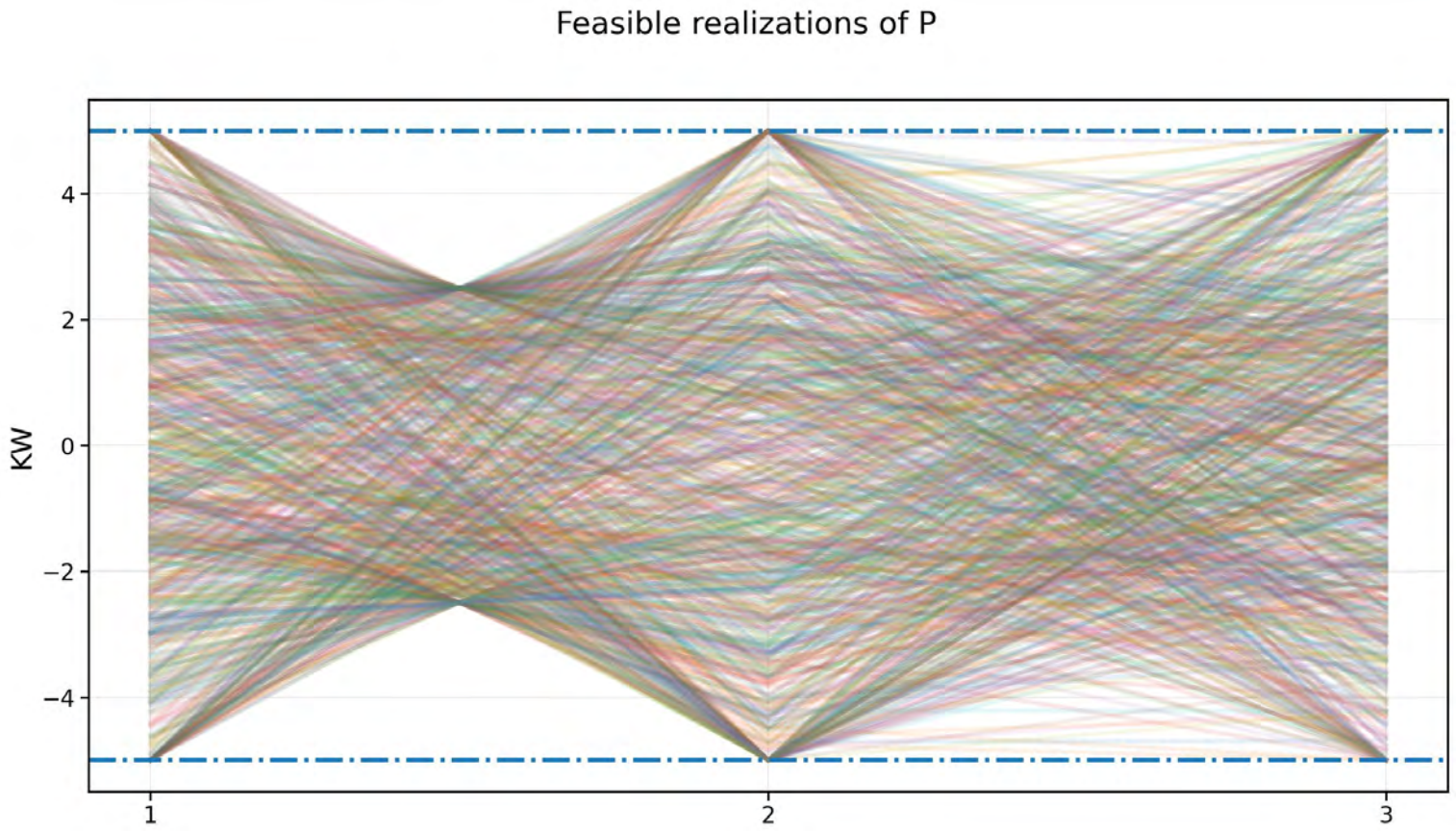
$$E_0 = 5.0$$
$$0 \leq E_0 + \boldsymbol{LP} \leq 10$$
$$-5 \leq \boldsymbol{P} \leq 5$$

Interactive scatter plot on Plotly

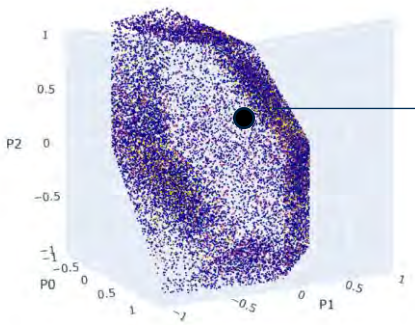


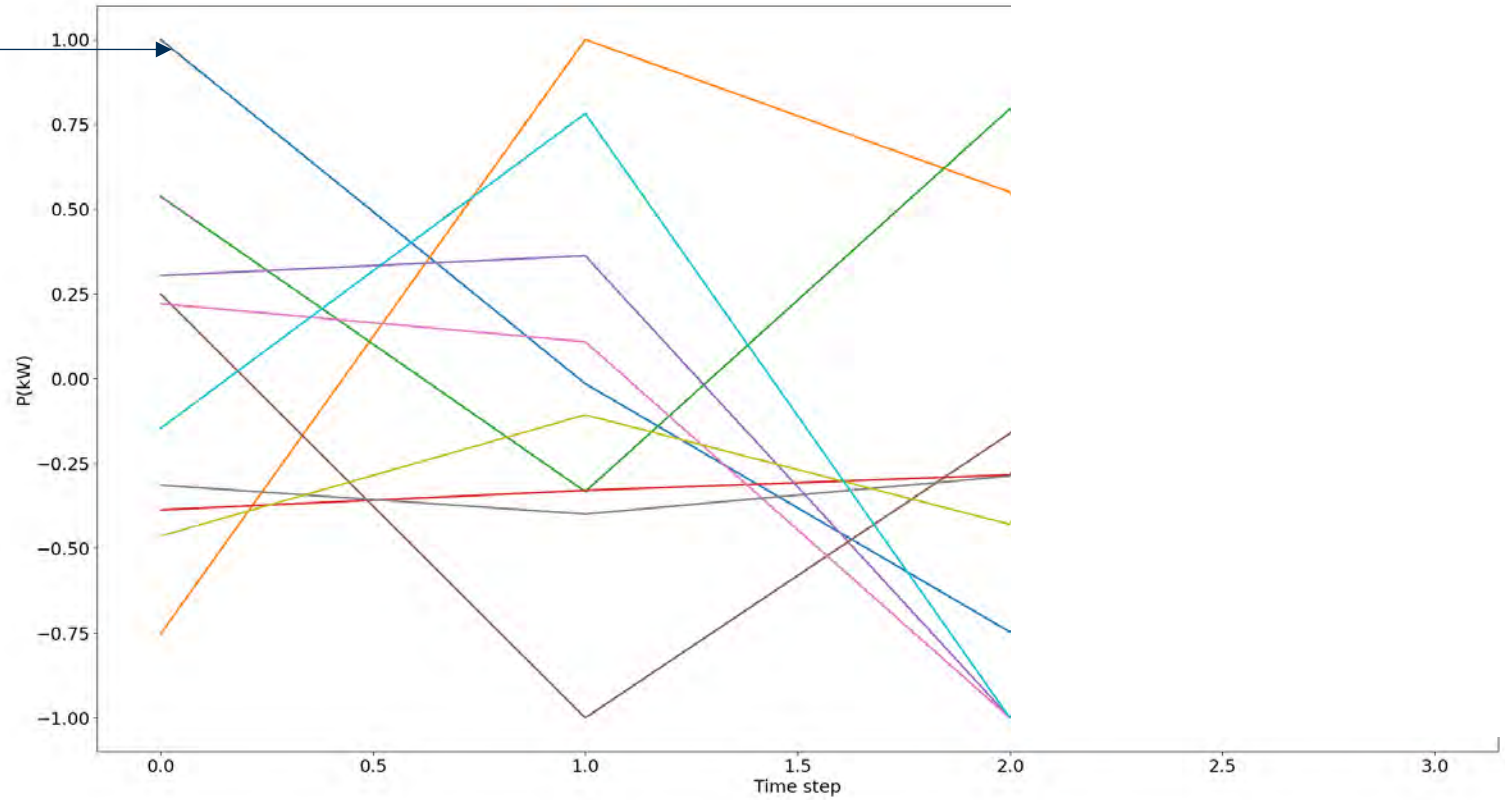
Time slot plot

On P space



A point in the polytope corresponds to one curve in P space





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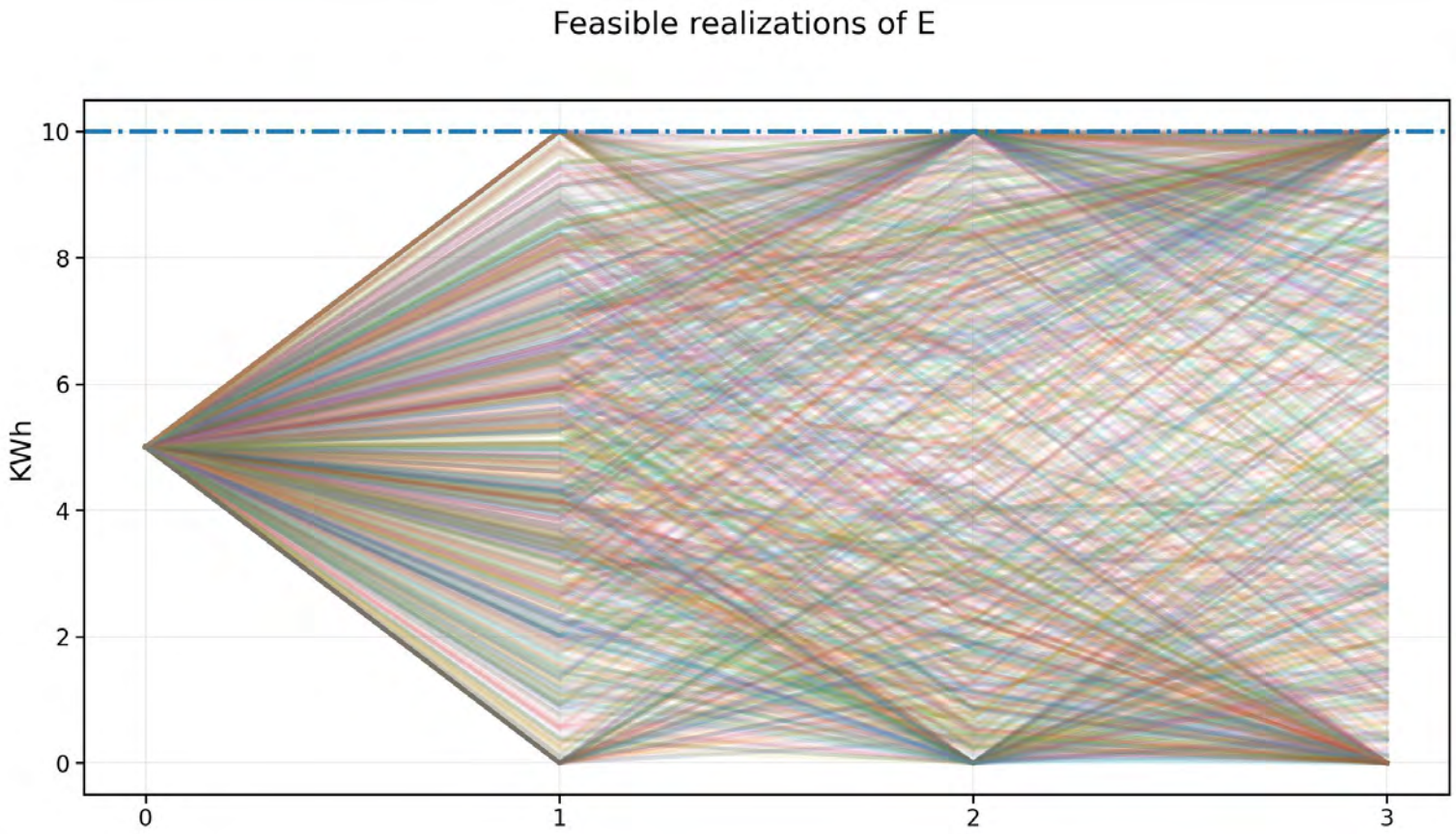
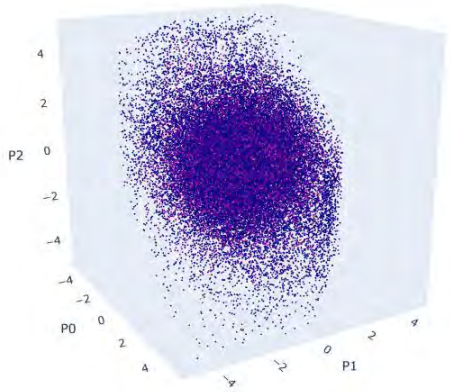
Time slot plot

On E space



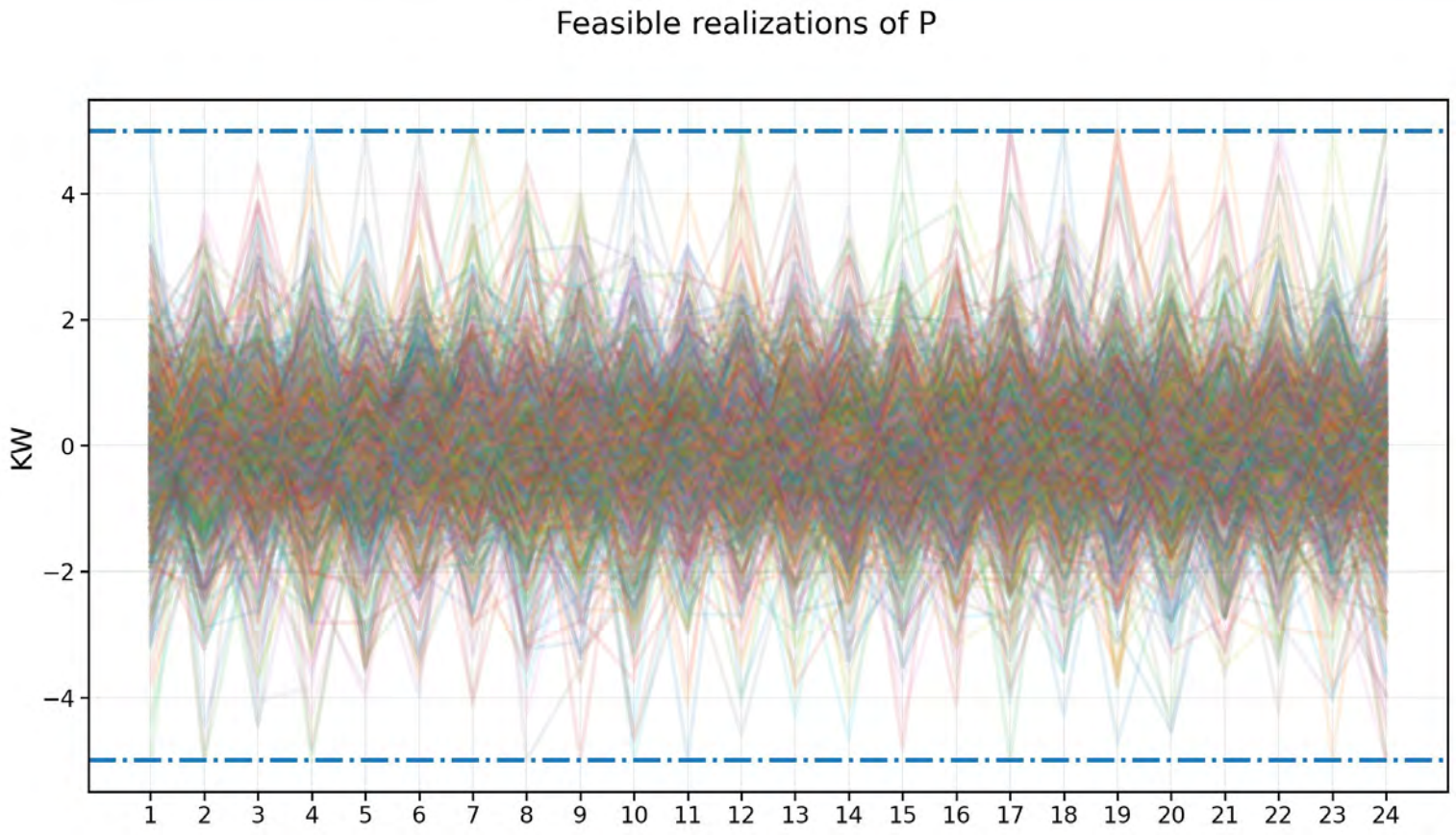
Now in a 24h range

Interactive scatter plot on Plotly



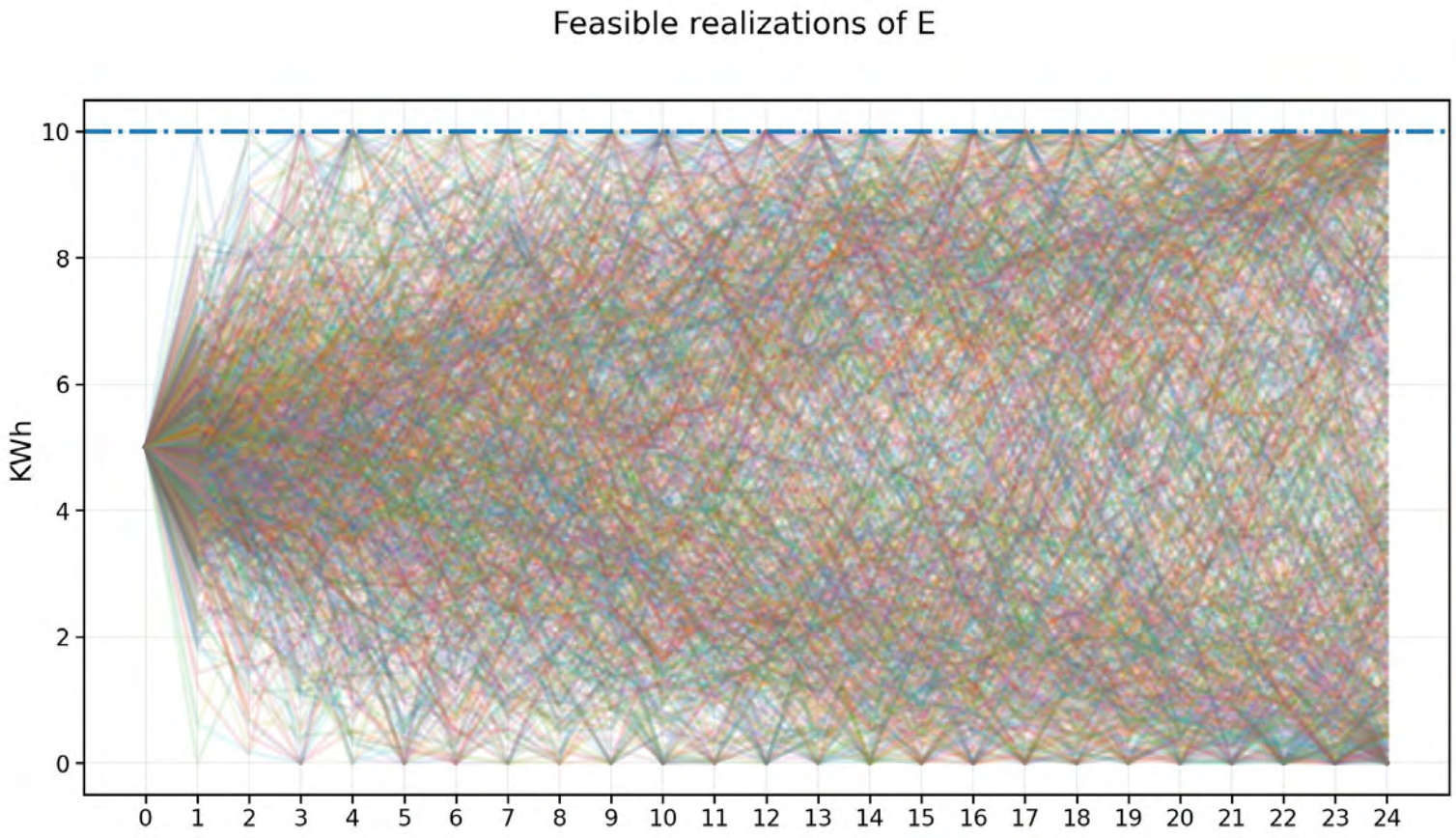
Time slot plot

On P space



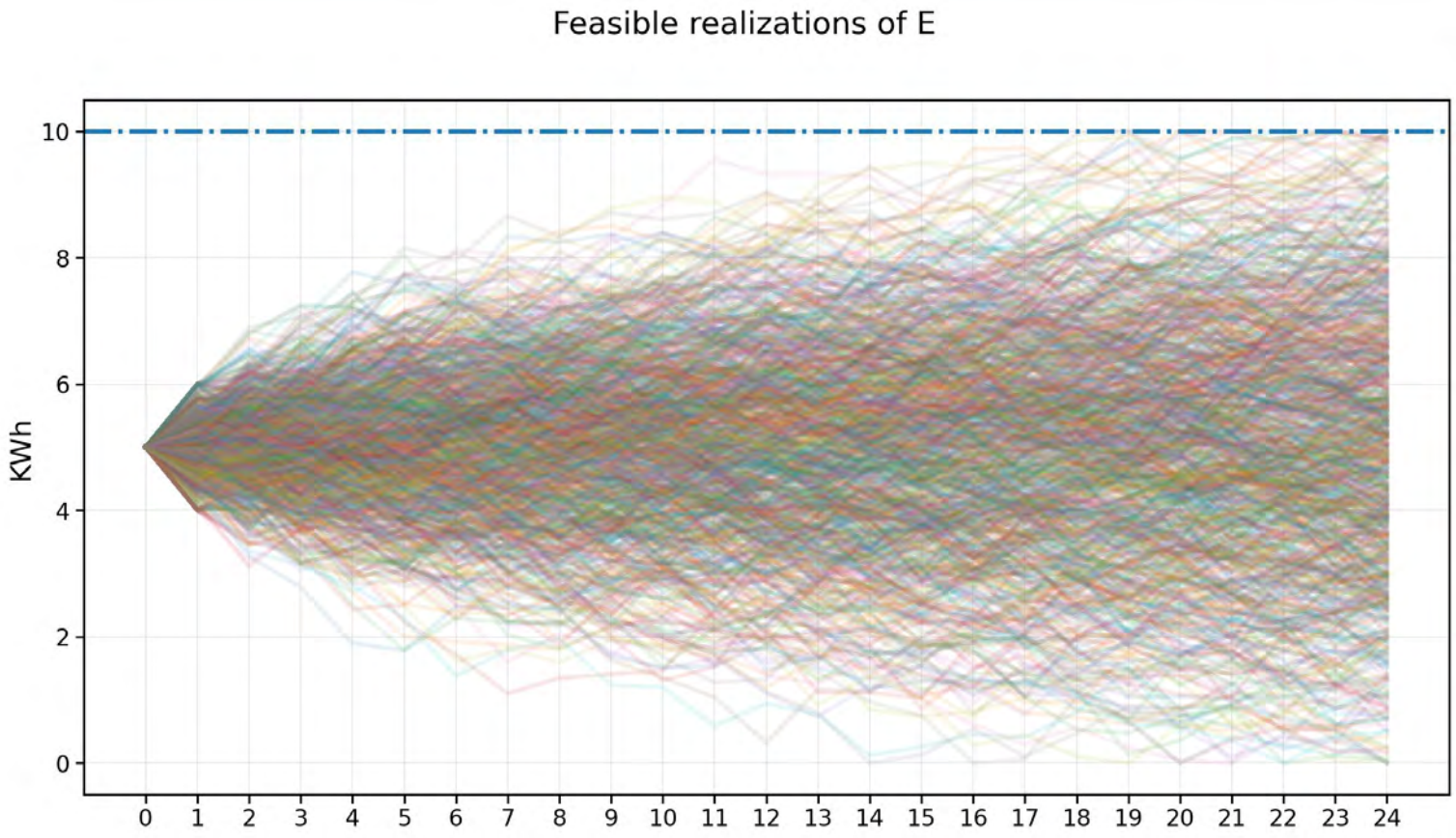
Time slot plot

On E space



Time slot plot

On P space, 1KW limits

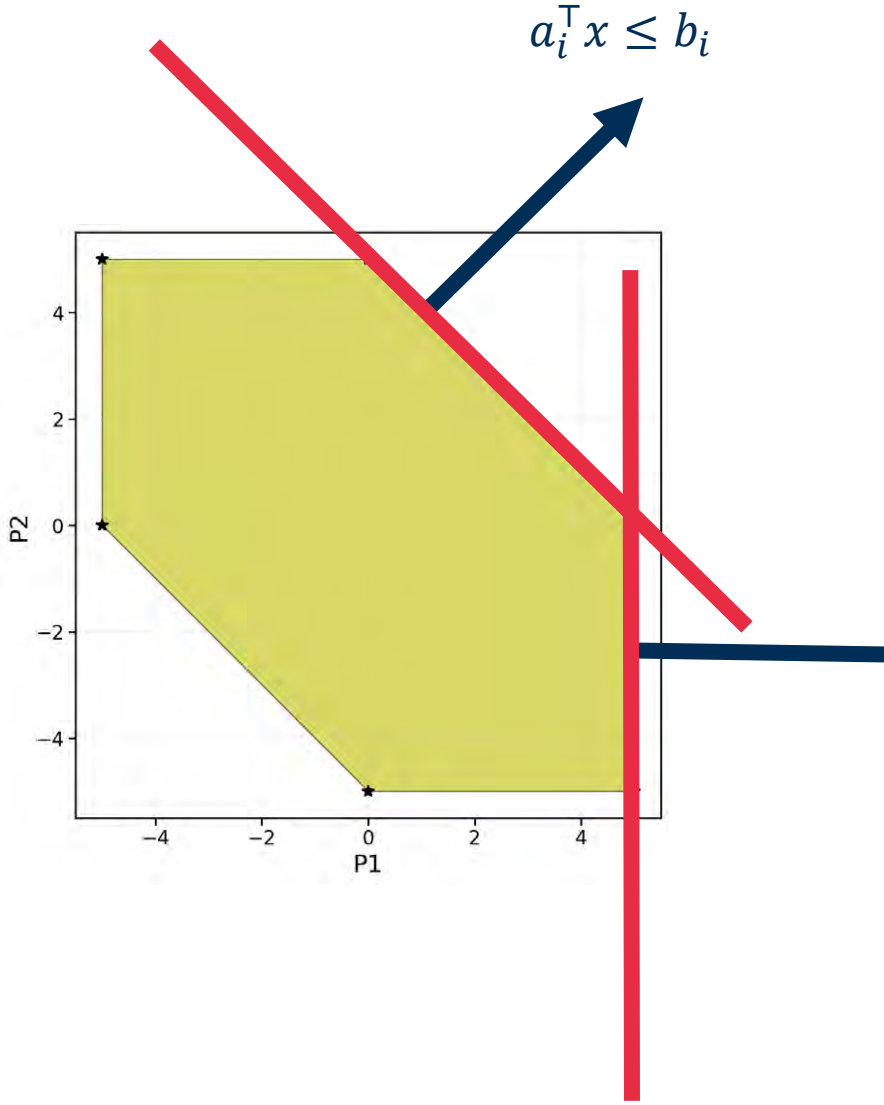


The fundamental results

There are two ways to represent a polytope

- $\mathcal{H} - rep$
  - $P = \{ x \in \mathbb{R}^N : Ax \leq b \}$
  - Half-plane representation, as seen in the presentation
  - Most natural inside optimizations

- We can represent the polytopes seen before as $Ax \leq b$
  - Each row represents a half-plane $a_i^\top x \leq b_i$
  - (A, b) representations are not unique

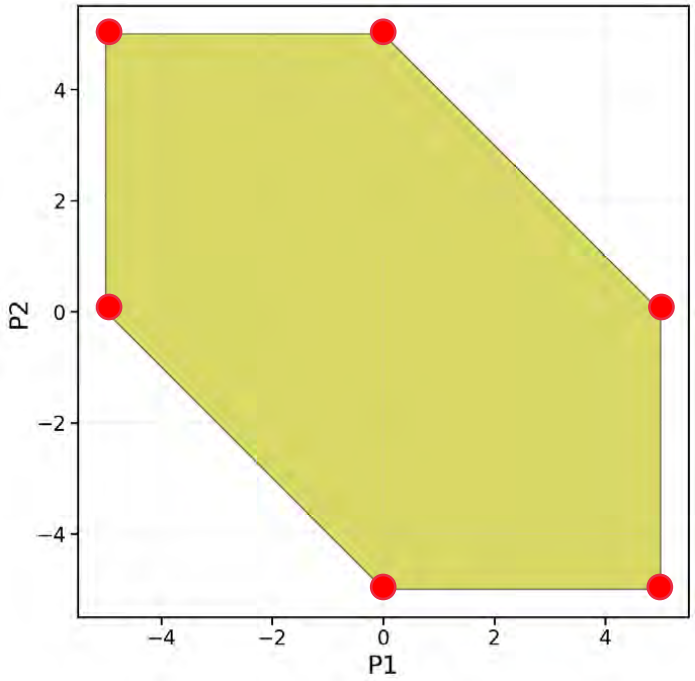


The fundamental results

There are two ways to represent a polytope (bounded)

- $\mathcal{V} - rep$
  - Simply consider the points in the region
  - Convexity: it is sufficient to consider only the extreme points
    - $P = conv(v_1, \dots, v_N)$

  - $v = \sum \lambda_i v_i, \lambda_i \geq 0, \sum \lambda_i = 1$



  - Take convex combinations of all vertices – easy to see in 2D or 3D

Minkowski–Weyl Theorem

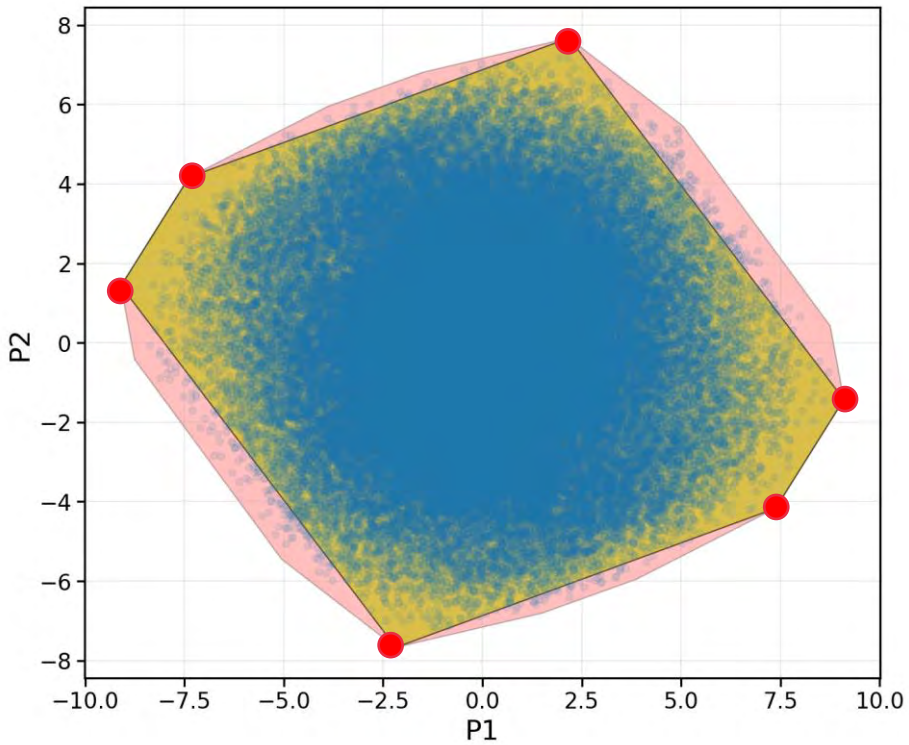
- A polytope (bounded polyhedron) can be written in any of these 2 ways

$$\mathcal{H} - rep \leftrightarrow \mathcal{V} - rep$$

- $\mathcal{H} - rep$ is usually more tractable

- Sometimes a vertex representation ($\mathcal{V} - rep$) is more convenient

- You can do ray-casting in $\mathcal{V} - rep$
  - Solve linear optimization problems
  - The convex combination of **any** number of extreme vertices will be inside the polytope



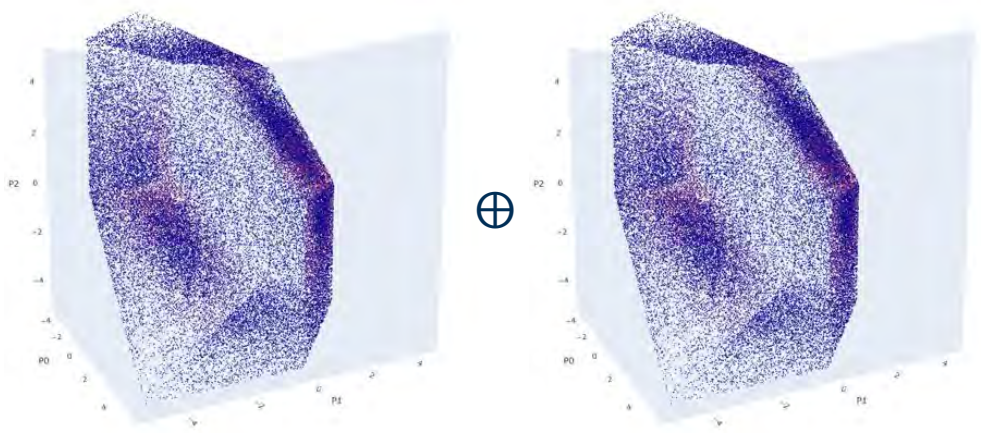
Aggregation of 2 loads

- It is very simple in $\boldsymbol{P}$ space:

$$\mathrm{P_{aggr}} = \{\mathrm{p_1} + \mathrm{p_2}, \qquad \text{for } \mathrm{p_1} \in \mathrm{P_1} \text{ and } \mathrm{p_2} \in \mathrm{P_2}\}$$

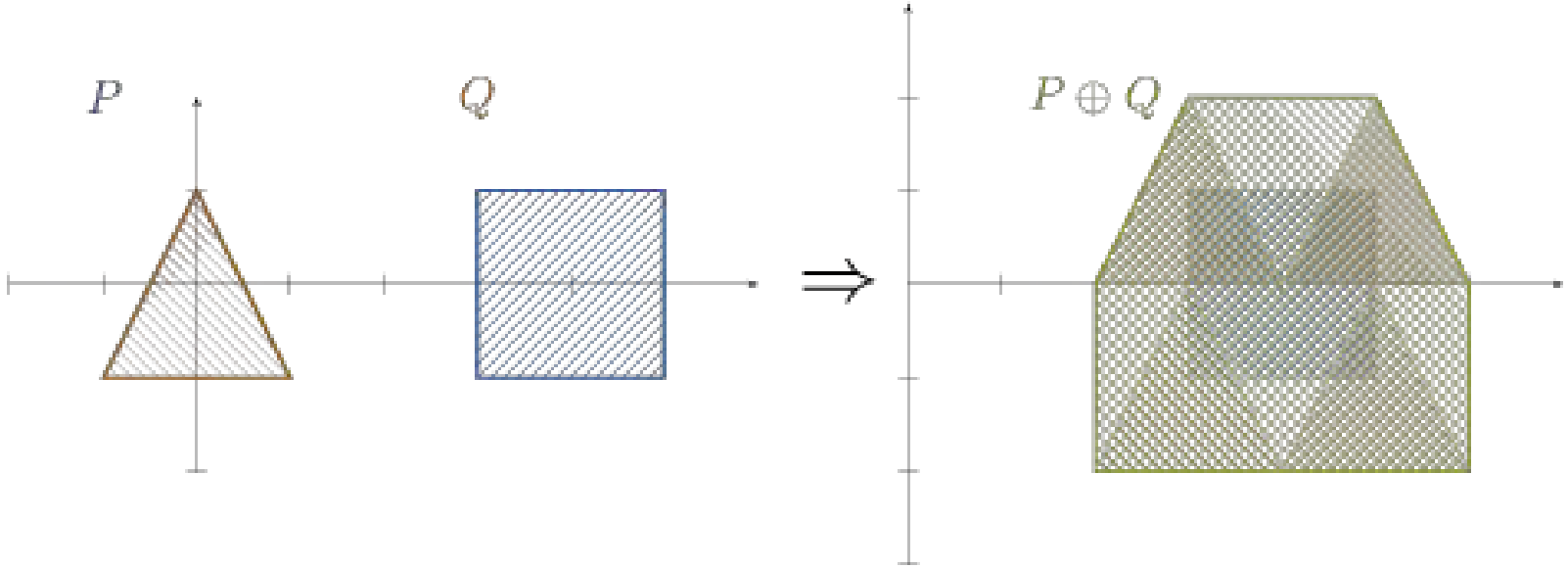
- Each p is a vector inside a polytope

- The result is also inside a polytope
  - But we don't have its representation!



Minkowski sum

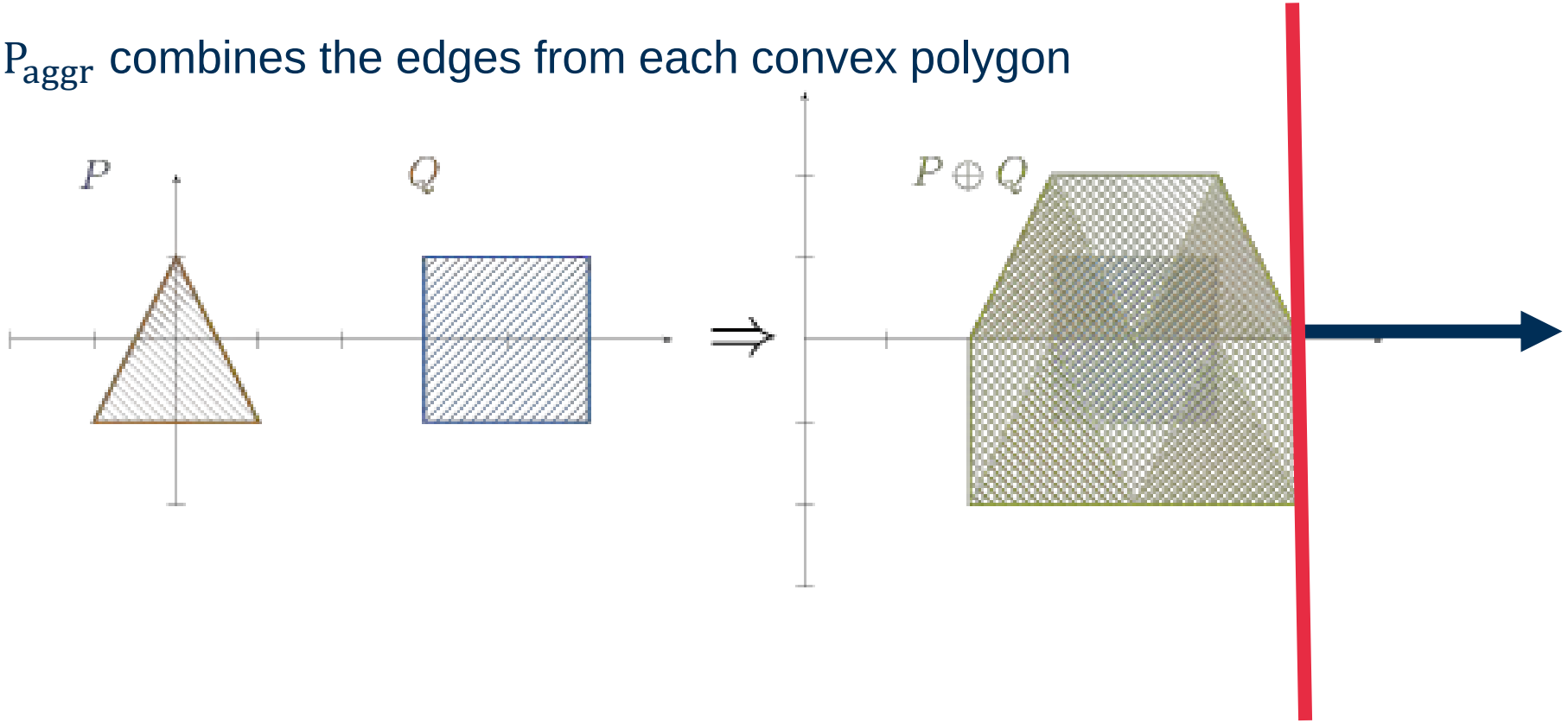
$$P_{aggr} = \{p_1 + p_2, \qquad \text{for } p_1 \in P_1 \text{ and } p_2 \in P_2\}$$



Minkowski sum

$$P_{aggr} = \{p_1 + p_2, \qquad \text{for } p_1 \in P_1 \text{ and } p_2 \in P_2\}$$

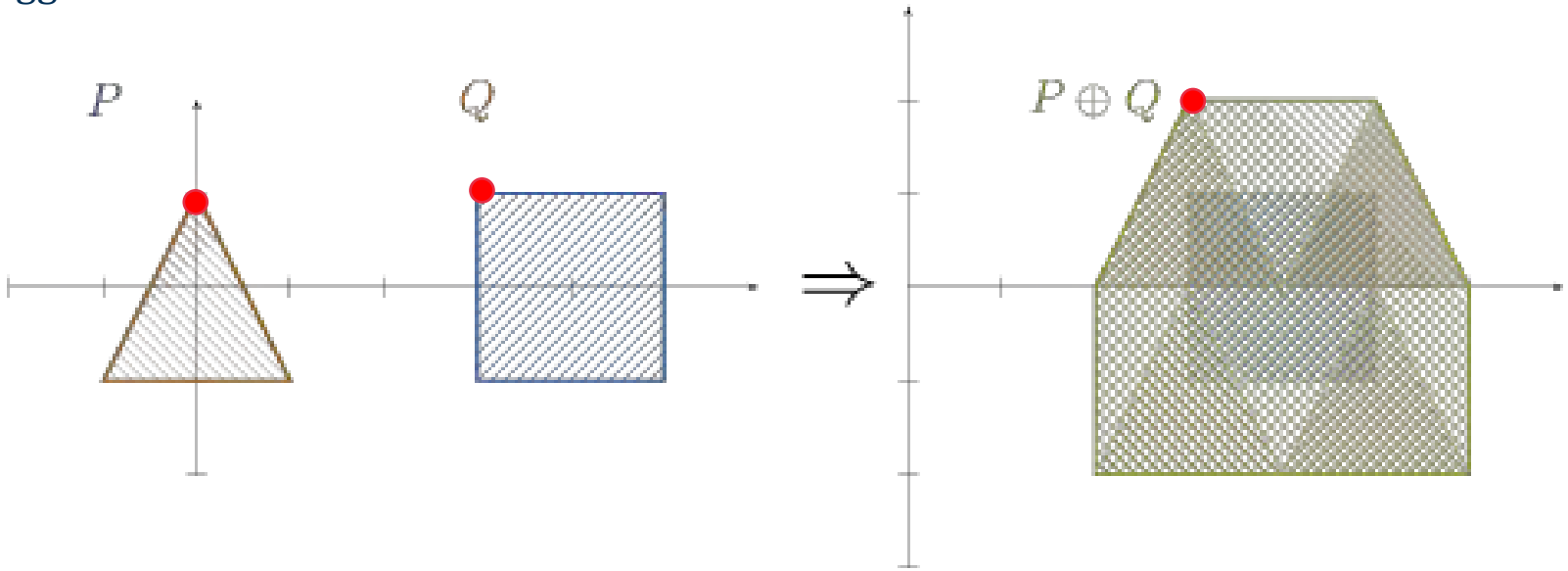
$\mathcal{H} - rep$: P_{aggr} combines the edges from each convex polygon



Minkowski sum

$$P_{aggr} = \{p_1 + p_2, \qquad \text{for } p_1 \in P_1 \text{ and } p_2 \in P_2\}$$

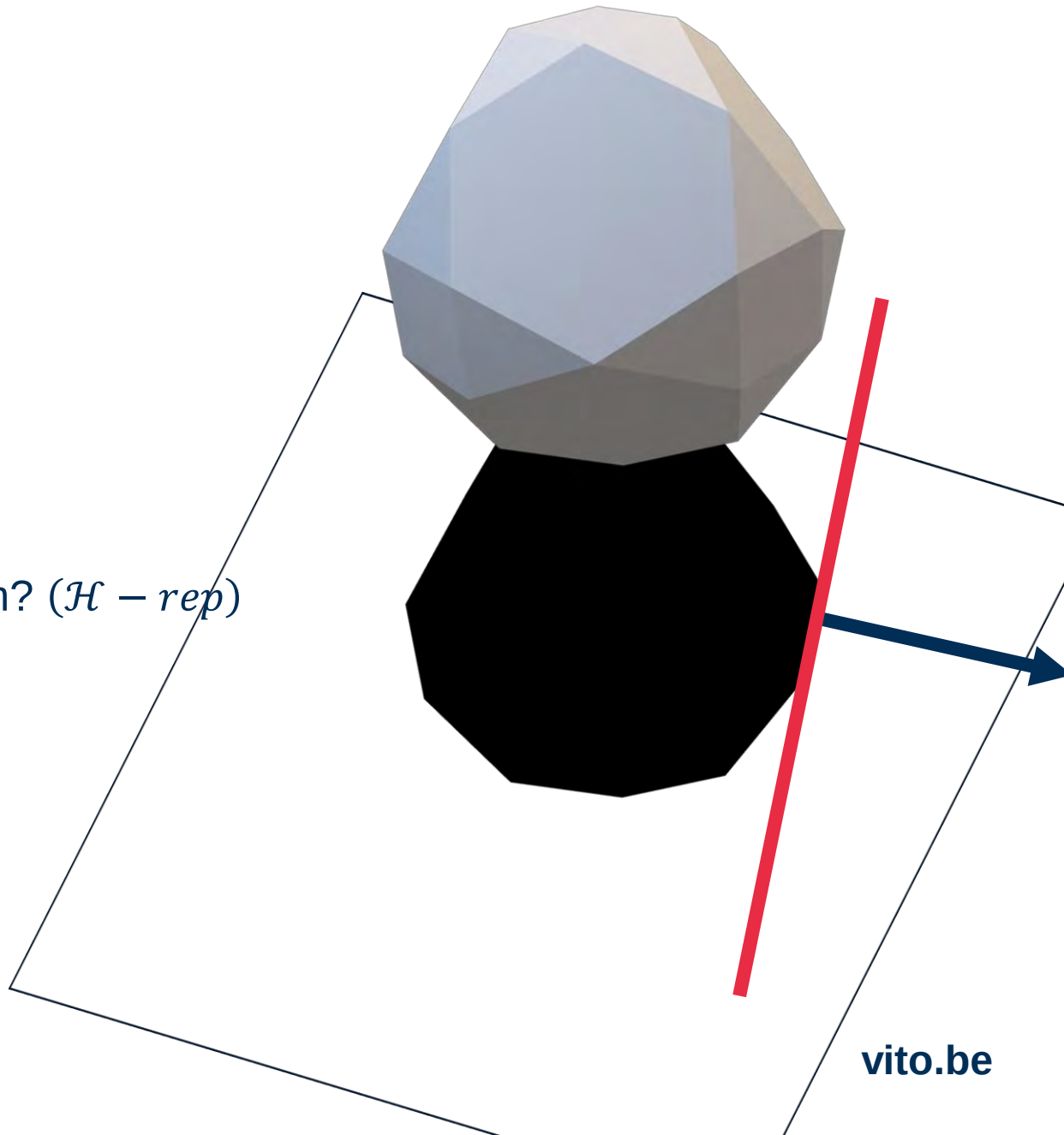
$\mathcal{V} - rep$: P_{aggr} combines the vertices from each convex polygon



Minkowski sum

In high dimensions

- $P_{aggr} = P_1 \oplus P_2$
- Define a variable $p_{aggr} = p_1 + p_2$
- Work in dimension 3x24, $(\mathrm{p}_1, \mathrm{p}_2, p_{aggr})$
- How to find a representation (A, b) for the sum? $(\mathcal{H} - rep)$

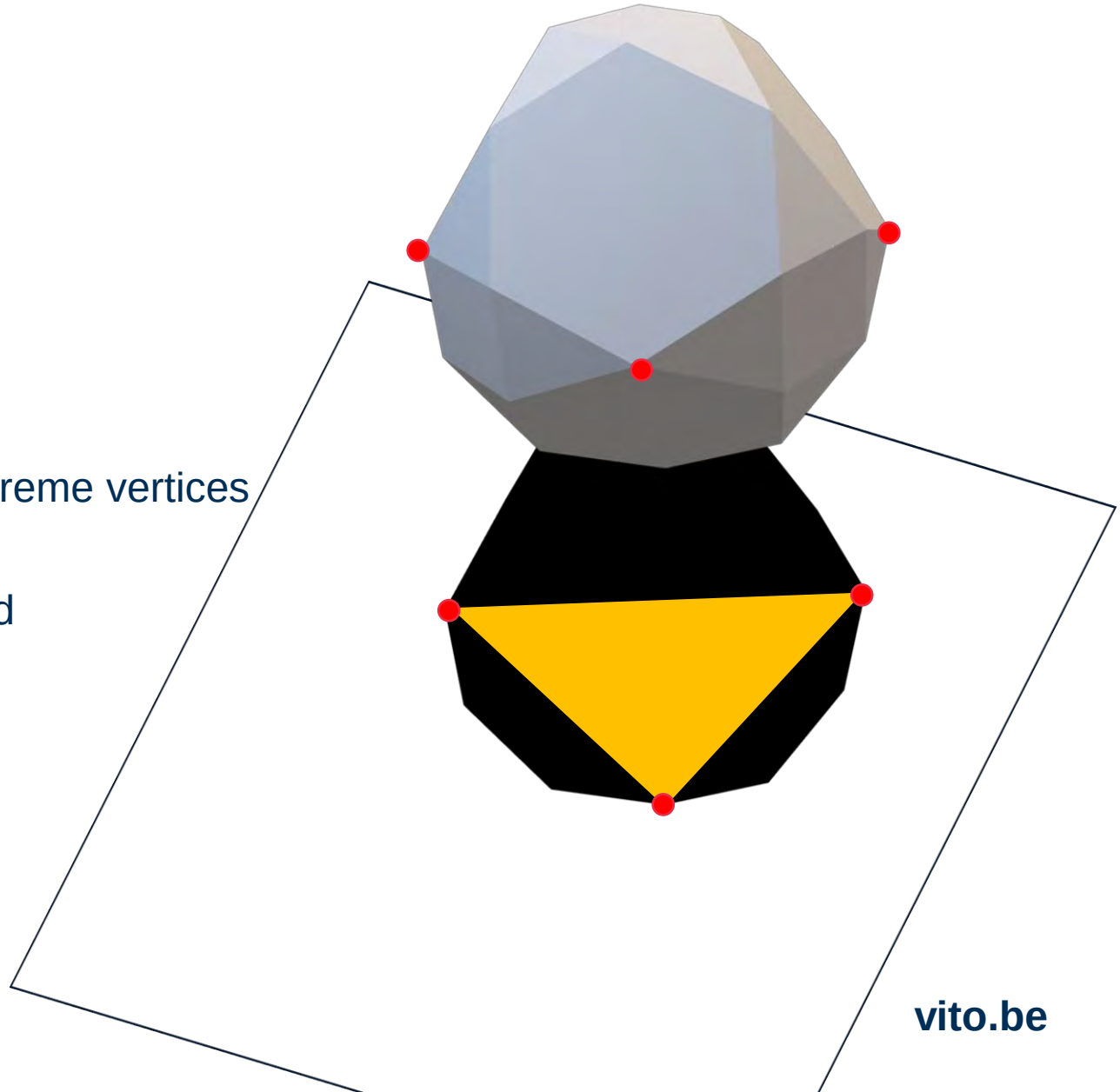


- **Projection** on the space of p_{aggr}
  - Fourier-Motzkin (similar to Gauss elimination)

Minkowski sum

- Polytope defined on $(\mathrm{p}_1, \mathrm{p}_2, p_{aggr})$
- Project on the plane $p_{aggr} = p1 + p2$

- Using the vertex representation $(\mathcal{V} - rep)$
  - It is easier - consider all combinations of extreme vertices
  - Vertices are easy to project
  - But still many combinations to be considered
    - And many combinations are not extreme
    - Convex hull



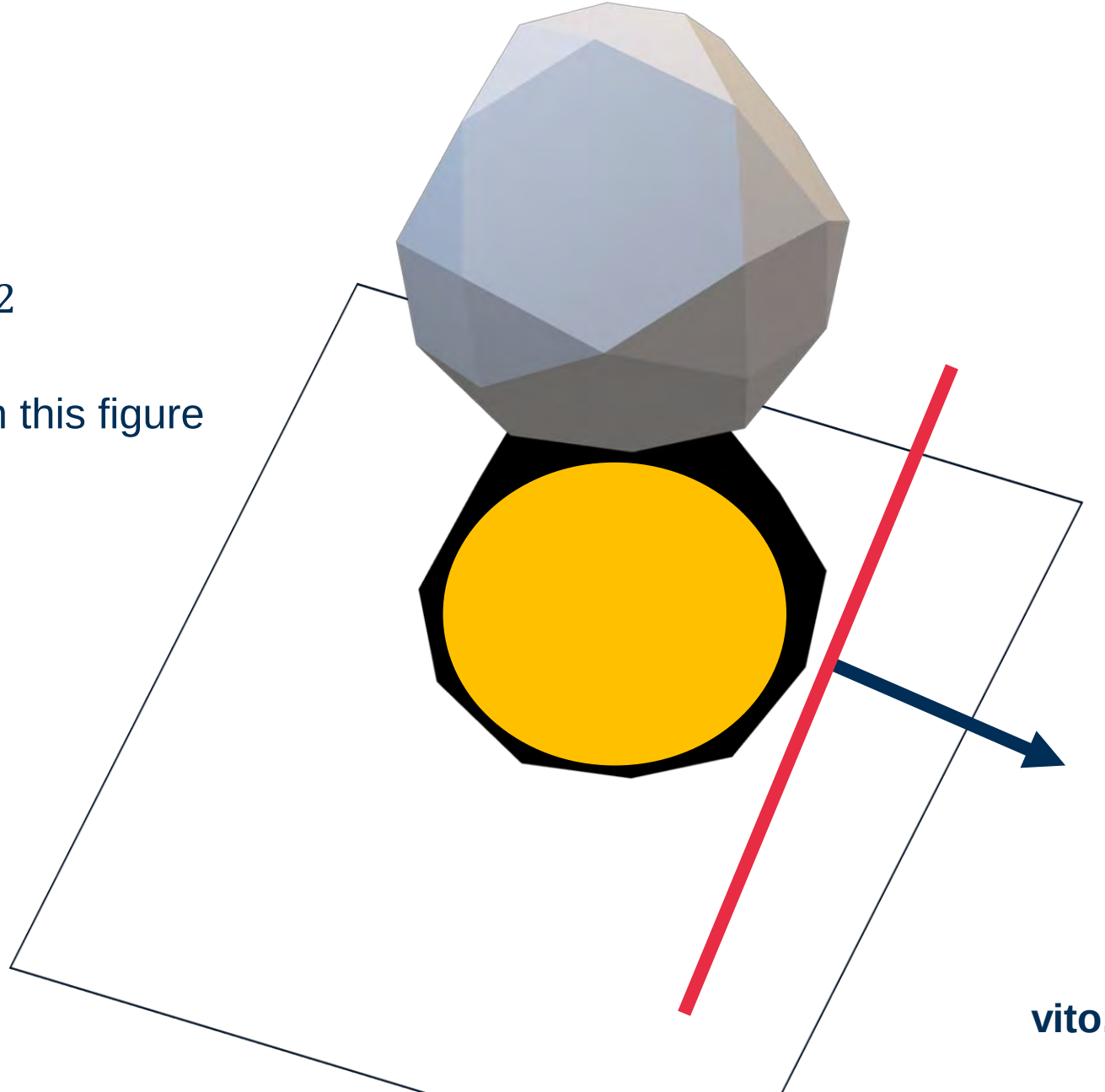
Minkowski sum

Tractable approaches

- Polytope defined on $(\mathrm{p}_1, \mathrm{p}_2, p_{aggr})$
- Project on the plane $p_{aggr} = p1 + p2$

- Find something about the shadow in this figure
  - Inner approximations
    - E.g. ray-casting $(\mathcal{V} - rep)$
  - Outer approximations

  - Containment ideas
    - Find largest box – very common
    - Find largest sphere
    - Zonotopes
    - λP for a tractable geometry P
      - homothety



Now consider many loads

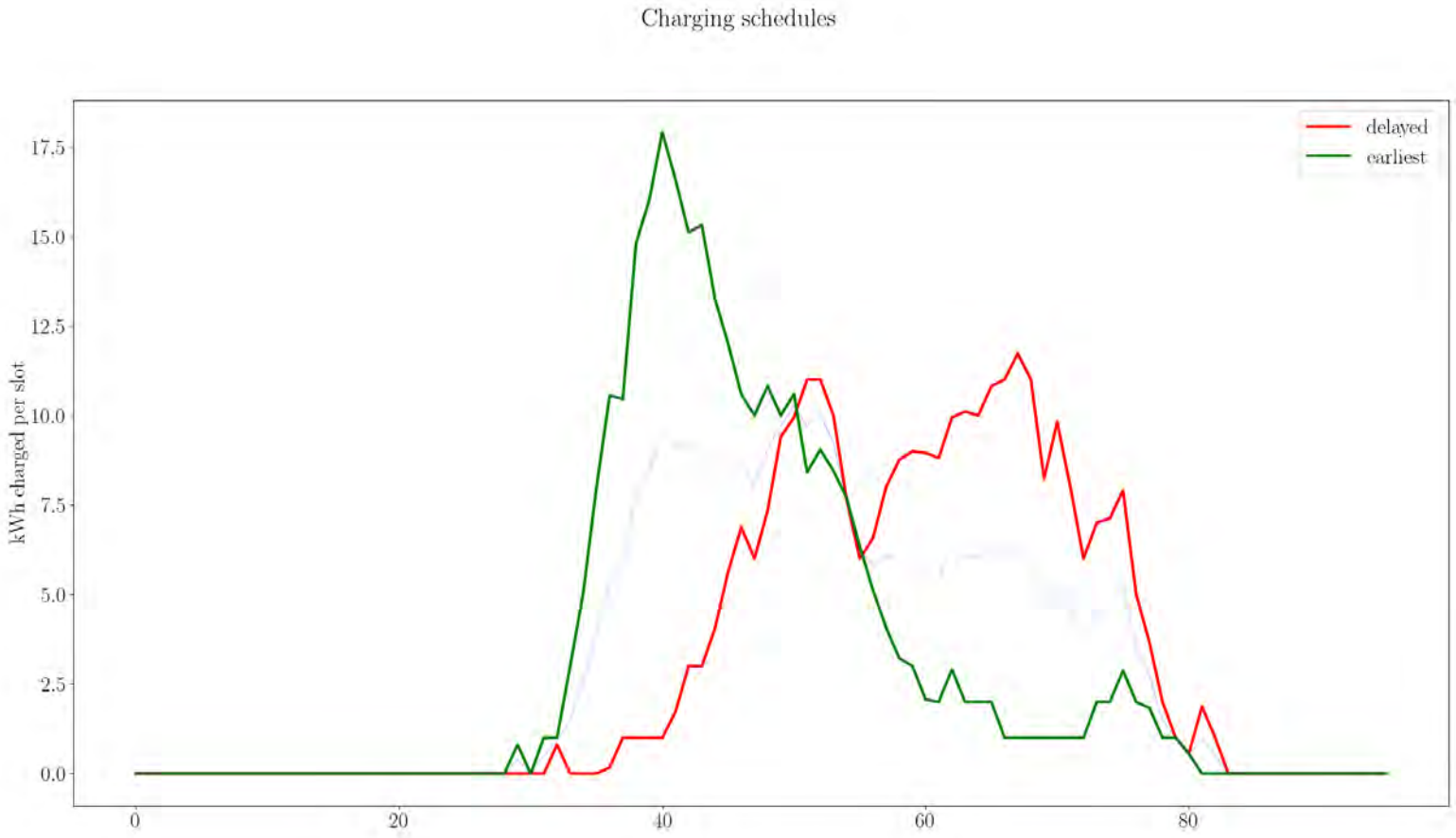
EV charging example

- EV = ‘battery on wheels’
  - Constrained to an arrival/departure schedule
    - On yellow slots:

$$0 \leq \mathrm{P_t} \leq P_i^{\max}$$

    - On dark slots:

  - Full charge before departure



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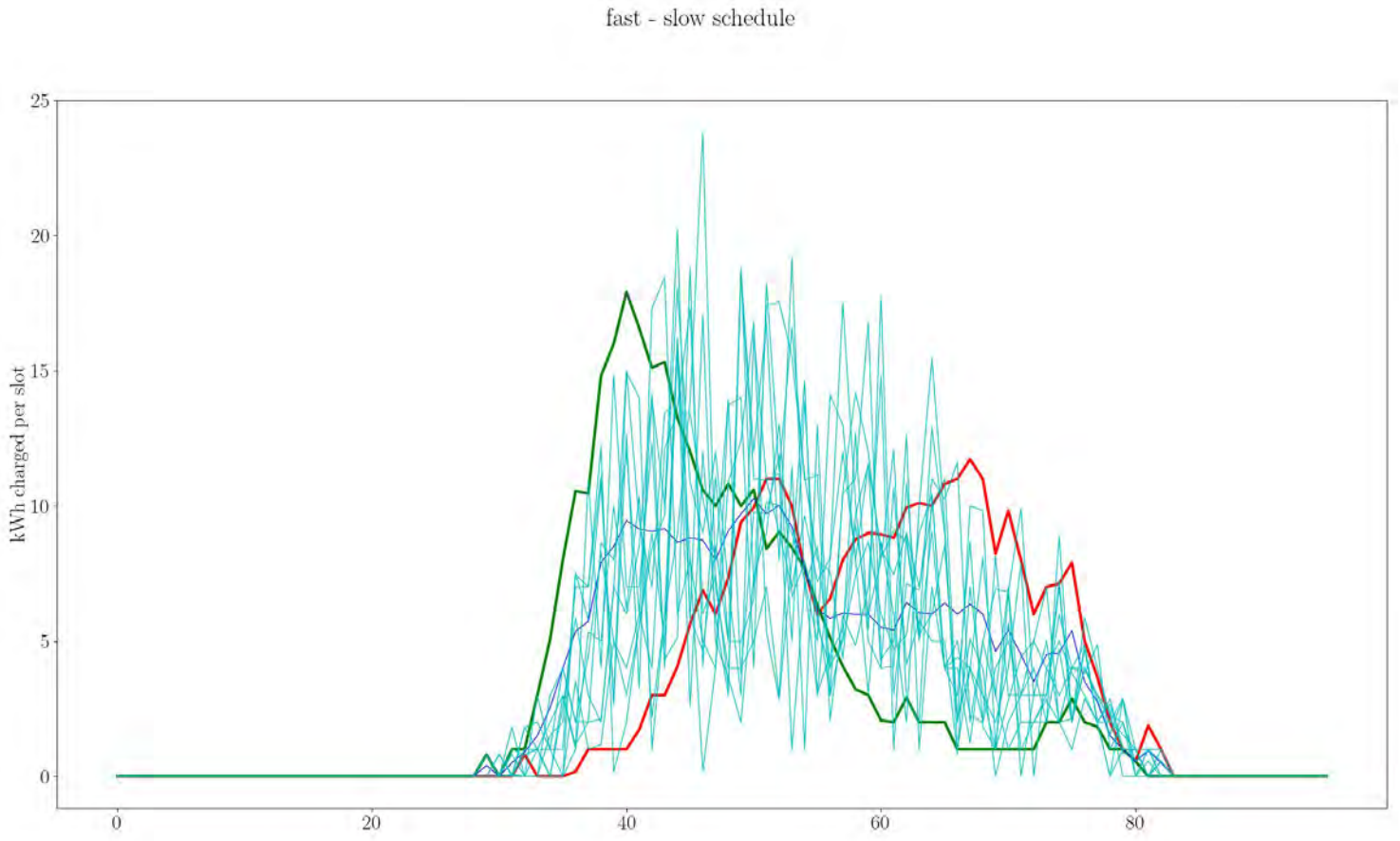
P per slot schedule



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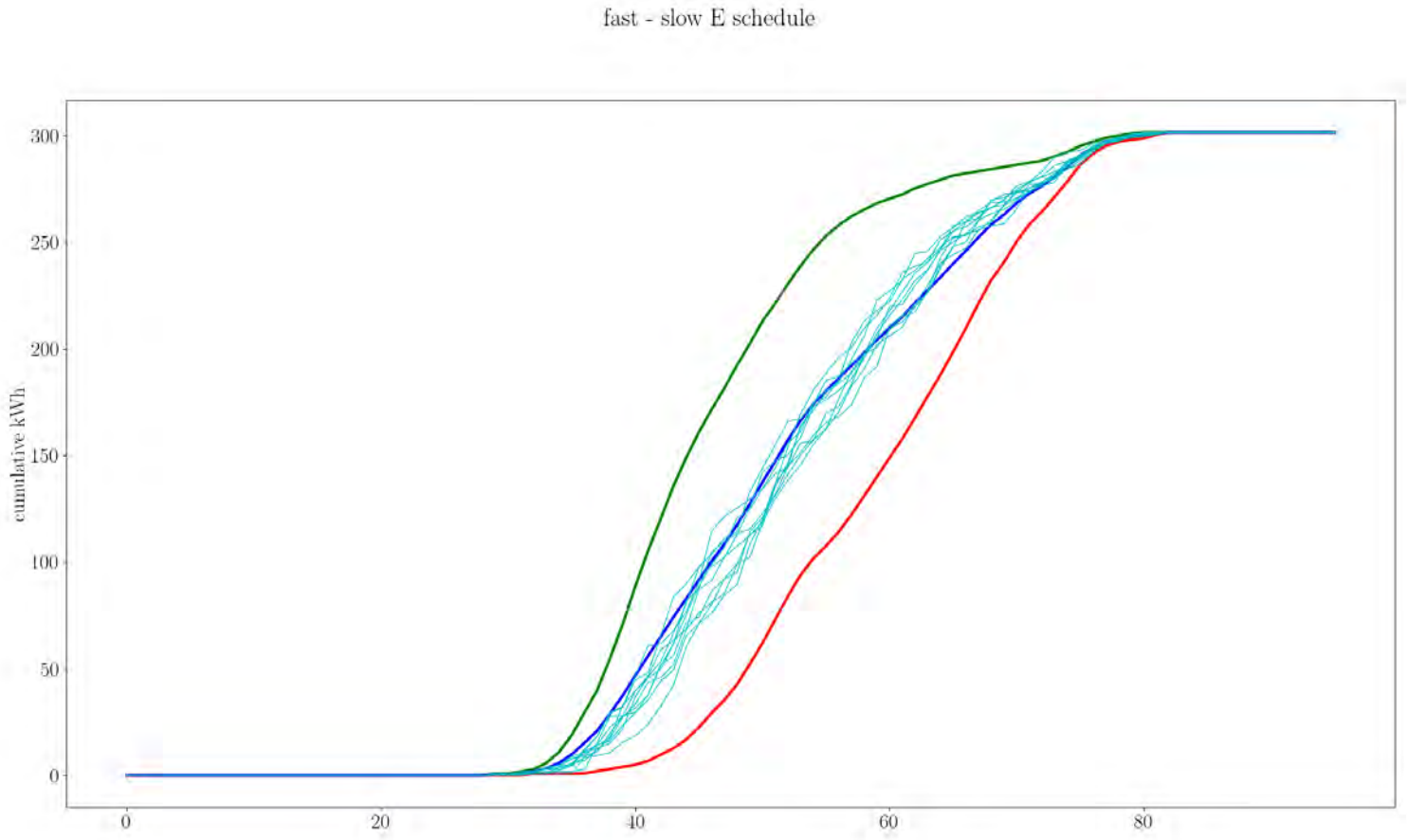
E / cumulative KWH schedule



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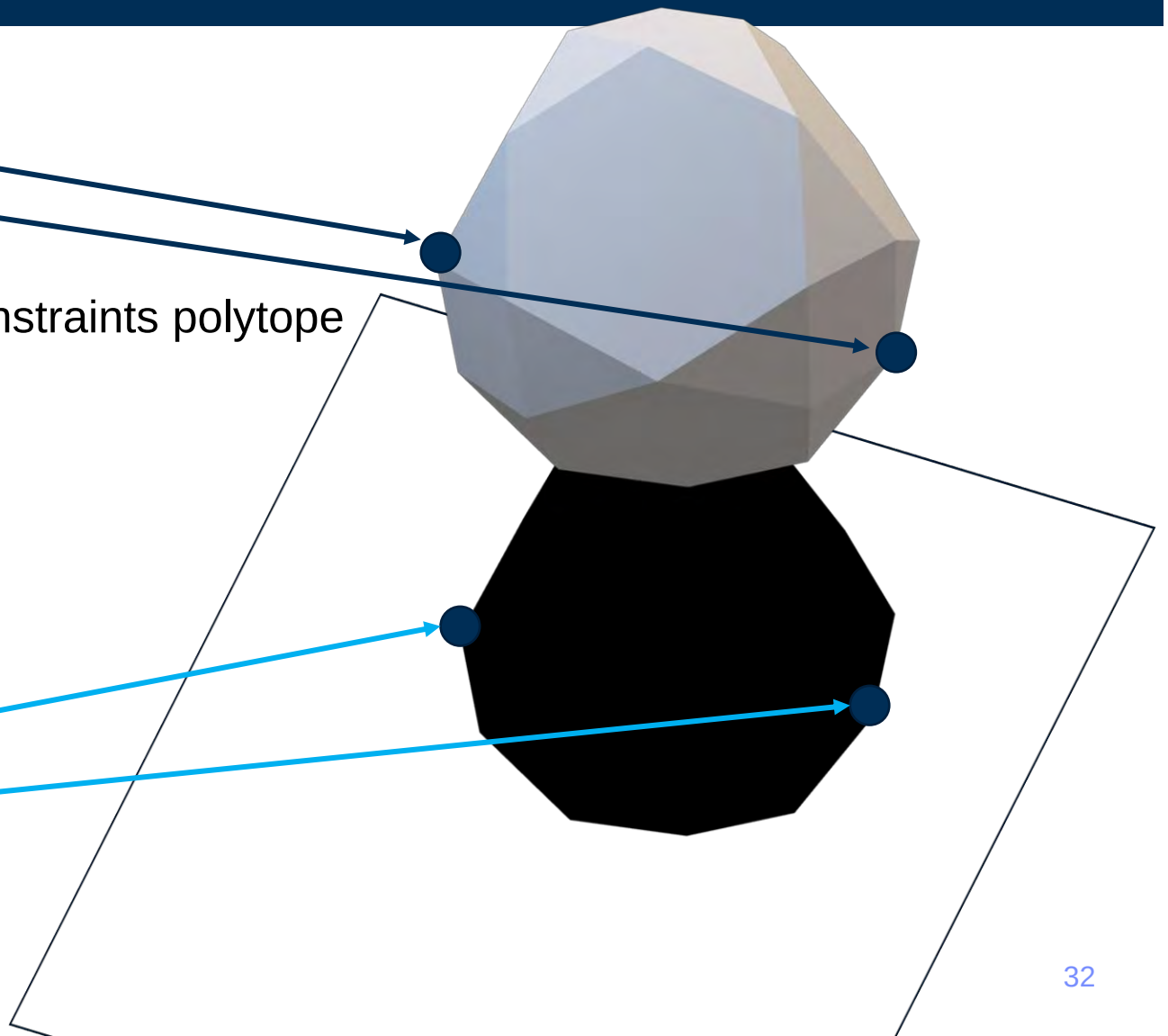
The upper/lower envelopes are feasible

- Earliest P^i schedules
- Latest P^i schedules

- The top is the full polytope
- The shadow represents the aggregated constraints polytope



Earliest aggregated schedule
Latest aggregated schedule

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Conclusions

- Interesting theoretical work on convex polytopes
- Powerful representation of flexibility and aggregation
- H-Rep matches what we use in optimizations
- But we must work around the limitations in high dimensions
  - Connects to the current STOA in literature

Thank you !

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